

Lab 3: Probability and Distributions

Instructor notes – Part 1: prep reading, Part 2: podium/circulating cheat sheet

This single file has two parts, both instructor-only (not a student handout). **Part 1** is read beforehand – an answer key and the reasoning behind each Act. **Part 2** (after the page break, large print) is the copy to have open while circulating during lab.

Schedule note: Labor Day cancels this week’s Monday meeting, so this lab runs with no prior lecture on Bayes’ rule – Wednesday’s lecture comes after, and the bestiary of distributions has moved to next Monday (Week 4). **Design note:** Act 1 is deliberately discovery-based rather than “apply the formula you were just taught” – students derive the PPV calculation themselves from plain counting before ever seeing it written as $P(H | D)$. **No laptops this lab** – all three Acts are pen-and-paper (a calculator is enough); Acts 2 and 3 use pre-rendered figures and a printed z-table instead of `dbinom()`/`dpois()`/`pnorm()`/`qnorm()`, since sampling/evaluating a named distribution isn’t, by itself, enough reason to open a laptop – that’s reserved for weeks with real algorithm-development content.

Suggested opening (~5 min, before releasing Act 1)

Keep this short – a full treatment of conditional probability and Bayes’ rule is Wednesday’s job, not today’s. Two rules are enough to unlock Act 1:

- **Addition:** probabilities of mutually exclusive (disjoint) outcomes add up (e.g., “flagged” splits into “flagged AND really the species” plus “flagged AND not the species” – no overlap, so these two pieces sum to the total flagged).
- **Multiplication / applying a rate:** a probability or rate applied to a count of events gives you the expected count in that subgroup (e.g., if 10 events are truly the species and the classifier catches 98% of them, expect about $10 \times 0.98 \approx 10$ flagged).

That’s it – don’t derive Bayes’ rule or use conditional-probability notation ($P(A | B)$) in this opening; Act 1 gets there on its own, and Wednesday’s lecture gives it a name.

Act 1: camera-trap classifier PPV (discovery-based)

Part 1 (counting). Same numbers as before ($P(\text{species}) = 0.0001$, sensitivity = 0.98, false positive rate = 0.01), $N = 100,000$.

```

N <- 1e5; p_S <- 1e-4; sens <- 0.98; fpr <- 0.01
species      <- N * p_S
not_species  <- N * (1 - p_S)
flag_S       <- species * sens
flag_notS    <- not_species * fpr
data.frame(row = c("Actually the species", "Not the species"),
           flagged = round(c(flag_S, flag_notS)),
           not_flagged = round(c(species - flag_S, not_species - flag_notS)),
           total = c(species, not_species))

```

```

##               row flagged not_flagged total
## 1 Actually the species      10          0     10
## 2      Not the species    1000     98990 99990

```

Answer key (1a-1b): Flagged/true-species ≈ 10 ; flagged/not-species $\approx 1,000$; total flagged $\approx 1,010$. PPV (1b) = $10/1010 \approx 0.0099 \approx 1\%$ – pure counting, no formula.

Answer key (1c): open-ended, but the number should read as “surprisingly low given how good the classifier sounded” – a single flag isn’t enough to trust; the field team should look for corroborating evidence (a second detection, a track, scat) before acting on one image.

Part 2 (formalize). Guided derivation:

```

# numerator: N * p_S * sens ; denominator: N*p_S*sens + N*(1-p_S)*fpr
# N cancels, leaving:
ppv_formula <- (p_S * sens) / (p_S * sens + (1 - p_S) * fpr)
ppv_formula

```

```
## [1] 0.009705853
```

Answer key (2a-2c): numerator = $N \times \text{prevalence} \times \text{sensitivity}$; denominator = $N \times \text{prevalence} \times \text{sensitivity} + N \times (1 - \text{prevalence}) \times \text{false positive rate}$. N cancels, leaving

$$\text{PPV} = \frac{\text{prevalence} \times \text{sensitivity}}{\text{prevalence} \times \text{sensitivity} + (1 - \text{prevalence}) \times \text{false positive rate}} \approx 0.0097$$

This is Bayes’ rule / PPV, arrived at without ever naming it – don’t name it for them either; that reveal is Wednesday’s job. If a group gets stuck on 2a, the hint is: “what did you multiply by 100,000 to get each cell of your table in Part 1? Do that again, but write N instead of 100,000.”

Part 3 (transfer – Site B). Prevalence = 0.01, same sensitivity and false positive rate.

```

p_S_B <- 0.01
ppv_B <- (p_S_B * sens) / (p_S_B * sens + (1 - p_S_B) * fpr)
ppv_B

```

```
## [1] 0.4974619
```

Answer key (3a): $PPV \approx 0.497 \approx 50\%$ – a dramatic jump from Site A’s $\approx 1\%$, same classifier. **(3b):** the intended takeaway is that prevalence dominates PPV every bit as much as the classifier’s own accuracy – a “good” classifier is only as useful as the base rate of what it’s looking for allows.

Watch for: groups who compute the fraction backwards ($P(S)$ instead of $P(F)$ in the denominator of 1b). A $PPV > 0.5$ at Site A is almost always the tell. In Part 2, watch for groups who try to cancel something other than N , or who get stuck because they’re trying to recall “the formula” instead of generalizing their own Part 1 numbers.

Act 2: binomial detections, and when Poisson is a good stand-in

$N = 8, p = 0.3, k = 3$; all figures below are pre-rendered in the handout (no code for students to run).

```
N <- 8; p <- 0.3; k <- 3
dbinom(k, N, p)
```

```
## [1] 0.2541218
```

```
which.max(dbinom(0:8, 8, 0.3)) - 1 # mode, 0-indexed
```

```
## [1] 2
```

Answer key: $P(X = 3) = 0.2541$; mode = 2.

Poisson comparison, part A ($N = 8, p = 0.3$ vs. $\lambda = 2.4$): binomial variance $Np(1 - p) = 1.68$; Poisson variance $\lambda = 2.4$ – mismatched, and the pre-rendered barplots look visibly different (Poisson has a fatter right tail).

Poisson comparison, part B ($N = 100, p = 0.024$ vs. $\lambda = 2.4$): binomial variance $= 100(0.024)(0.976) = 2.3424$ – close to 2.4, and the barplots should look nearly identical. Rule of thumb (2d): *Poisson is a good stand-in for binomial when N is large and p is small* (so $(1 - p) \approx 1$ and $Np(1 - p) \approx Np = \lambda$).

Watch for: groups who conclude Poisson is “just always fine” from Scenario B alone without noticing Scenario A disagreed – push them back to the variance mismatch in Scenario A as the reason it isn’t universal.

Act 3: continuous body mass ($\mu = 4.5, \sigma = 0.6$), via z-table

```
mu <- 4.5; sigma <- 0.6
z <- (5.5 - mu) / sigma
z
```

```
## [1] 1.666667
```

```

1 - pnorm(round(z, 2))          # table-precision version of 1 - Phi(z)

## [1] 0.04745968

1 - pnorm(5.5, mu, sigma)      # exact, for comparison

## [1] 0.04779035

mu + 1.28 * sigma              # table-lookup version of the 90th percentile

## [1] 5.268

qnorm(0.9, mu, sigma)          # exact, for comparison

## [1] 5.268931

```

Answer key: $z = (5.5 - 4.5)/0.6 = 1.67$ (table row 1.6, column .07); $\Phi(1.67) \approx 0.9525$, so $P(\text{mass} > 5.5) \approx 0.0475$ ($\approx 5\%$; the exact value via `pnorm` is 0.0478 – table rounding accounts for the small difference). For 3c, the table’s closest entry to 0.90 is $\Phi(1.28) = 0.8997$, giving a cutoff of $4.5 + 1.28(0.6) \approx 5.27$ kg (exact value via `qnorm` is 5.27 kg – matches).

Watch for: 1b/1c (density vs. probability) is the conceptual crux of Part 1 – a group that answers “ $P(\text{mass} = 4.5)$ is very small but not zero” hasn’t fully separated density from probability yet. Point them back at the shaded-area picture from 1a: a single point has zero *width*, hence zero area, hence zero probability, regardless of how tall the curve is there. In Part 3, watch for groups who forget to convert the z-score back to a body mass in 3c (the table gives z , not x).

PART 2 – CHEAT SHEET (circulating)

Copy for circulating during lab. Full reasoning and code: see Part 1 above.

Opening (~5 min)

- Two rules only: **addition** (disjoint outcomes sum) and **multiplication/rate-application** ($\text{rate} \times \text{count} = \text{expected count in a subgroup}$)
- Don't derive Bayes' rule or use $P(A | B)$ notation – Act 1 gets there on its own; Wednesday names it

Act 1 – classifier PPV (discovery)

- Part 1 (count, $N = 100\text{k}$): flagged/true ≈ 10 , flagged/false $\approx 1,000$, total flagged $\approx 1,010 \rightarrow \text{PPV} \approx 1\%$
- Part 2 hint if stuck: “what did you multiply by 100,000 in Part 1? Do that again with N instead”
- Part 2 answer: $\text{PPV} = \frac{\text{prev} \times \text{sens}}{\text{prev} \times \text{sens} + (1 - \text{prev}) \times \text{fpr}} \approx 0.0097$
- Part 3 (Site B, prevalence = 0.01): $\text{PPV} \approx 50\%$ – prevalence dominates
- **Watch for:** $\text{PPV} > 0.5$ at Site A usually means $P(S)/P(F)$ flipped

Act 2 – binomial vs. Poisson

- $N = 8, p = 0.3, k = 3$: hand calc = 0.2541; mode = 2
- Var mismatch ($N = 8, p = 0.3$): binom 1.68 vs. Pois 2.4 – **don't match**
- Var match ($N = 100, p = 0.024$): binom 2.3424 vs. Pois 2.4 – **do match**
- Rule of thumb: Poisson $\approx$ binomial when N large, p small

Act 3 – continuous mass, via z-table

- $z = (5.5 - 4.5)/0.6 = 1.67 \rightarrow \Phi(1.67) \approx 0.9525 \rightarrow P(\text{mass} > 5.5) \approx 0.048$
- Table's closest entry to 0.90 is $z \approx 1.28 \rightarrow \text{cutoff} \approx 4.5 + 1.28(0.6) = 5.27$ kg
- **Watch for:** “small but nonzero” answers to $P(X = 4.5)$ (push back to zero width $\rightarrow$ zero area); forgetting to convert z back to x in 3c